

Low Vitamin D Status: Definition, Prevalence, Consequences and Correction

Vitamin D is obtained from cutaneous production when 7-dehydrocholesterol is converted to vitamin D₃ (cholecalciferol) by ultraviolet B radiation or by oral intake of vitamin D₂ (ergocalciferol) and D₃. An individual's vitamin D status is best evaluated by measuring the circulating 25-hydroxyvitamin D [25(OH)D] concentration. Though controversy surrounds the definition of low vitamin D status, there is increasing agreement that the optimal circulating 25(OH)D level should be ~30-32 ng/ml or above. Using this definition, it has been estimated that approximately three quarters of all adults in the United States are low. Classically, low vitamin D status has skeletal consequences such as osteomalacia/rickets. More recently, associations between low vitamin D status and increased risk for various non-skeletal morbidities have been recognized; whether all of these associations are causally related to low vitamin D status remains to be determined. To achieve optimal vitamin D status, daily intakes of at least 1000 IU or more of vitamin D are required. The risk of toxicity with "high" amounts of vitamin D intake is low. Substantial between-individual variability exists in response to the same administered vitamin D dose. When to monitor 25(OH)D levels has received little attention. Supplementation with vitamin D₃ may be preferable to vitamin D₂.